

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399034
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Kapil; Swarnakshi

Routing of an autonomous vehicle in a flood condition

Abstract

Systems and techniques are provided for routing an autonomous vehicle (AV) in a flood condition. An example method can include collecting sensor data from one or more AVs, wherein the sensor data includes measurements associated with a level of precipitation in one or more locations; receiving map data descriptive of an elevation of ground at the one or more locations; determining that the level of precipitation in the one or more locations relative to the elevation of ground at the one or more locations is higher than a threshold; and transmitting an alert signal to the one or more AVs, the alert signal notifying of a flood condition at the one or more locations.

Inventors:	Kapil; Swarnakshi (Sunnyvale, CA)
Applicant:	GM Cruise Holdings LLC (San Francisco, CA)
Family ID:	1000008780267
Assignee:	GM Cruise Holdings LLC (San Francisco, CA)
Appl. No.:	18/094647
Filed:	January 09, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240230370 A1	Jul. 11, 2024

Publication Classification

Int. Cl.: G01C21/00 (20060101); B60W40/02 (20060101)

U.S. Cl.:

CPC G01C21/3885 (20200801); B60W40/02 (20130101); G01C21/3841 (20200801); B60W2420/408 (20240101); B60W2555/20 (20200201); B60W2556/40 (20200201);

Field of Classification Search

CPC: B60W (40/02); B60W (2420/408); B60W (2555/20); B60W (2556/40); B60W (2556/45); G01C (21/00); G01C (21/3461); G01C (21/3694); G01C (21/3841); G01C (21/3885)

USPC: 701/400

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9110196	12/2014	Urmson	N/A	G01S 17/42
9207323	12/2014	Zhu	N/A	G01S 17/86
9499172	12/2015	Urmson	N/A	B60W 40/06
9632210	12/2016	Zhu	N/A	G01S 17/87
10255782	12/2018	Ghannam	N/A	G08B 21/20
11046271	12/2020	Lerner	N/A	B60W 50/0098
11441916	12/2021	Konrardy	N/A	G01C 21/343
11449946	12/2021	Gutsell	N/A	G06N 20/00
11594017	12/2022	Gupta	N/A	G01S 15/931
11959745	12/2023	Gray, Jr.	N/A	H04W 4/38
2014/0081507	12/2013	Urmson	701/28	B60W 40/06
2014/0081573	12/2013	Urmson	356/600	G01S 17/42
2014/0307247	12/2013	Zhu	342/54	B60W 40/064
2014/0336935	12/2013	Zhu	702/3	G01N 21/4738
2017/0025002	12/2016	Claudel	N/A	G08G 1/096844
2018/0004211	12/2017	Grimm	N/A	G05D 1/0066
2018/0348753	12/2017	Ortiz Gonzalez	N/A	B60W 40/02
2019/0316309	12/2018	Wani	N/A	G06F 3/0484
2019/0376807	12/2018	Iagnemma	N/A	G01C 21/3461
2020/0130622	12/2019	Lerner	N/A	G01C 21/3697
2021/0213976	12/2020	Salter	N/A	G05D 1/0291
2022/0157178	12/2021	Grace et al.	N/A	N/A
2022/0164911	12/2021	Stumpf	N/A	H04W 4/024
2022/0188866	12/2021	Farmer	N/A	G06Q 10/02
2022/0198196	12/2021	Beaurepaire	N/A	G06V 40/103

2023/0100961	12/2022	Pan	701/24	G01C 21/3629
2023/0182742	12/2022	Han	701/23	B60W 40/02
2023/0236317	12/2022	Fina	342/26B	G01S 7/412
2024/0107635	12/2023	Gu	N/A	G01S 7/497
2024/0111057	12/2023	Woll	N/A	G01S 7/4802
2024/0118424	12/2023	Shane	N/A	G01S 17/931
2024/0149881	12/2023	Villegas	N/A	B60W 60/001
2024/0166213	12/2023	Ahmed	N/A	B60W 40/068
2024/0230370	12/2023	Kapil	N/A	G08G 1/0133

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2021229238	12/2020	AU	G06V 20/188
104768822	12/2014	CN	B60W 30/18
109849923	12/2018	CN	B60W 30/18009
110058240	12/2018	CN	B60W 30/09
118535871	12/2023	CN	N/A
118799767	12/2023	CN	N/A
119672566	12/2024	CN	N/A
119737932	12/2024	CN	N/A
119780000	12/2024	CN	N/A
102021123436	12/2022	DE	G06Q 10/02
4064238	12/2021	EP	B60W 50/0097
20210090720	12/2020	KR	B60W 40/02
2686159	12/2018	RU	B64C 39/02
WO-2022229706	12/2021	WO	N/A

OTHER PUBLICATIONS

E. B. Panganiban and J. C. Dela Cruz, "Rain water level information with flood warning system using flat clustering predictive technique," TENCON 2017-2017 IEEE Region 10 Conference, Penang, Malaysia, 2017, pp. 727-732, doi: 10.1109/TENCON.2017.8227956.

(<https://ieeexplore.ieee.org/document/8227956>) (Year: 2017). cited by examiner

Lo SW, Wu JH, Lin FP, Hsu CH. Visual Sensing for Urban Flood Monitoring. Sensors (Basel).

Aug. 14, 2015; 15(8):20006-29. doi: 10.3390/s150820006. PMID: 26287201; PMCID:

PMC4570407. (Year: 2015). cited by examiner

Panganiban, et al., "Rain water level information with flood warning system using flat clustering predictive technique," TENCON 2017-2017 IEEE Region 10 Conference, Penang, Malaysia, 2017, pp. 727-732, doi: 10.1109/TENCON.2017.8227956.

(<https://ieeexplore.ieee.org/document/8227956>) (Year: 2017). cited by examiner

Lo, et al. Visual Sensing for Urban Flood Monitoring. Sensors (Basel). Aug. 14, 2015;

15(8):20006-29. doi: 10.3390/s150820006. PMID: 26287201; PMCID: PMC4570407.

(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4570407/>) (Year: 2015). cited by examiner

Background/Summary

BACKGROUND

1. Technical Field

(1) The present disclosure generally relates to autonomous vehicles and, more specifically, to routing of an autonomous vehicle in a flood condition.

2. Introduction

(2) An autonomous vehicle is a motorized vehicle that can navigate without a human driver. An exemplary autonomous vehicle can include various sensors, such as a camera sensor, a light detection and ranging (LIDAR) sensor, and a radio detection and ranging (RADAR) sensor, amongst others. The sensors collect data and measurements that the autonomous vehicle can use for operations such as navigation. The sensors can provide the data and measurements to an internal computing system of the autonomous vehicle, which can use the data and measurements to control a mechanical system of the autonomous vehicle, such as a vehicle propulsion system, a braking system, or a steering system. Typically, the sensors are mounted at fixed locations on the autonomous vehicles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The various advantages and features of the present technology will become apparent by reference to specific implementations illustrated in the appended drawings. A person of ordinary skill in the art will understand that these drawings only show some examples of the present technology and would not limit the scope of the present technology to these examples. Furthermore, the skilled artisan will appreciate the principles of the present technology as described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1 illustrates an example system environment that can be used to facilitate autonomous vehicle (AV) dispatch and operations, according to some examples of the present disclosure;

(3) FIG. 2 illustrates a diagram that includes an AV detecting a flood condition, according to some examples of the present disclosure;

(4) FIG. 3 illustrates an example system for routing and/or dispatching an AV in a flood condition, according to some examples of the present disclosure;

(5) FIG. 4 illustrates an example process for routing an AV in a flood condition, according to some examples of the present disclosure; and

(6) FIG. 5 illustrates an example processor-based system with which some aspects of the subject technology can be implemented.

DETAILED DESCRIPTION

(7) The detailed description set forth below is intended as a description of various configurations of the subject technology and is not intended to represent the only configurations in which the subject technology can be practiced. The appended drawings are incorporated herein and constitute a part of the detailed description. The detailed description includes specific details for the purpose of providing a more thorough understanding of the subject technology. However, it will be clear and

apparent that the subject technology is not limited to the specific details set forth herein and may be practiced without these details. In some instances, structures and components are shown in block diagram form in order to avoid obscuring the concepts of the subject technology.

(8) One aspect of the present technology is the gathering and use of data available from various sources to improve quality and experience. The present disclosure contemplates that in some instances, this gathered data may include personal information. The present disclosure contemplates that the entities involved with such personal information respect and value privacy policies and practices.

(9) As previously explained, autonomous vehicles (AVs) can include various sensors, such as a camera sensor, a Light Detection and Ranging (LiDAR) sensor, a Radio Detection and Ranging (RADAR) sensor, an audio sensor, amongst others, which the AVs can use to collect data and measurements that the AVs can use for operations such as navigation. The sensors can provide the data and measurements to an internal computing system of the AV, which can use the data and measurements to control a mechanical system of the AV, such as a vehicle propulsion system, a braking system, or a steering system.

(10) Heavy rainfalls associated with floods may be predicted and detected using satellites, lightning observing systems, and rain gauges. Based on such predictions and detection, a flood warning can be provided to the public through public warning systems (e.g., radio or television broadcasters, wireless emergency alerts, etc.). For example, a flash flood warning can be issued to alert the public that a flash flood is imminent or occurring in the warned area or region. However, even a small road segment within the same region can have varying elevations that would result in a different road condition in a flood event. Also, floods can be progressing at a rapid pace in some locations. Therefore, it can be difficult and complicated to get accurate predictions and detailed real-time information on a potential, imminent, or occurring flood condition. Accurate and timely detection of any potential or progressing flood condition (e.g., detection of a change in a water level on road) is important to provide real-time warnings and route vehicles (and passengers) to a safe spot to ensure the safe operation of the vehicles and the safety of passengers.

(11) Described herein are systems, apparatuses, processes (also referred to as methods), and computer-readable media (collectively referred to as “systems and techniques”) for routing and/or dispatching an AV in a flood condition. In some examples, the systems and techniques described herein can predict and/or determine a potential, imminent, or occurring flood condition on road in real time by leveraging data from various sources such as sensor data collected from one or more AVs, map data, absorbability data, historical data, or any applicable data associated with weather or meteorological data. Further, the systems and techniques described herein can, based on the predictions and detection of a flood condition, route/re-route an AV to avoid a location or a spot that has a high level of precipitation that may affect or endanger the operation of the AV. For example, the systems and techniques described herein can transmit an alert signal to an AV to notify a flood condition, for example, in a visualization of the flood condition on a map that defines a zone or an area in which an AV should not travel.

(12) In some aspects, the systems and techniques described herein can transmit an alert signal to a user who may be associated with an AV. For example, an alert signal to a user can provide an updated map in real time that shows a zone or an area that is affected by a flood event (e.g., a visualization of the updated map on a display of a wireless device associated with the user). Further, in response to a determination that a user is located within a threshold distance from a flooded area, the systems and techniques described herein can dispatch an AV to help the user evacuate and move to a safe spot.

(13) FIG. 1 is a diagram illustrating an example autonomous vehicle (AV) environment **100**, according to some examples of the present disclosure. One of ordinary skill in the art will understand that, for the AV environment **100** and any system discussed in the present disclosure, there can be additional or fewer components in similar or alternative configurations. The

illustrations and examples provided in the present disclosure are for conciseness and clarity. Other examples may include different numbers and/or types of elements, but one of ordinary skill in the art will appreciate that such variations do not depart from the scope of the present disclosure.

(14) In this example, the AV environment **100** includes an AV **102**, a data center **150**, and a client computing device **170**. The AV **102**, the data center **150**, and the client computing device **170** can communicate with one another over one or more networks (not shown), such as a public network (e.g., the Internet, an Infrastructure as a Service (IaaS) network, a Platform as a Service (PaaS) network, a Software as a Service (SaaS) network, other Cloud Service Provider (CSP) network, etc.), a private network (e.g., a Local Area Network (LAN), a private cloud, a Virtual Private Network (VPN), etc.), and/or a hybrid network (e.g., a multi-cloud or hybrid cloud network, etc.).

(15) The AV **102** can navigate roadways without a human driver based on sensor signals generated by multiple sensor systems **104**, **106**, and **108**. The sensor systems **104-108** can include one or more types of sensors and can be arranged about the AV **102**. For instance, the sensor systems **104-108** can include Inertial Measurement Units (IMUs), cameras (e.g., still image cameras, video cameras, etc.), light sensors (e.g., LIDAR systems, ambient light sensors, infrared sensors, etc.), RADAR systems, GPS receivers, audio sensors (e.g., microphones, Sound Navigation and Ranging (SONAR) systems, ultrasonic sensors, etc.), engine sensors, speedometers, tachometers, odometers, altimeters, tilt sensors, impact sensors, airbag sensors, seat occupancy sensors, open/closed door sensors, tire pressure sensors, rain sensors, and so forth. For example, the sensor system **104** can be a camera system, the sensor system **106** can be a LIDAR system, and the sensor system **108** can be a RADAR system. Other examples may include any other number and type of sensors.

(16) The AV **102** can also include several mechanical systems that can be used to maneuver or operate the AV **102**. For instance, the mechanical systems can include a vehicle propulsion system **130**, a braking system **132**, a steering system **134**, a safety system **136**, and a cabin system **138**, among other systems. The vehicle propulsion system **130** can include an electric motor, an internal combustion engine, or both. The braking system **132** can include an engine brake, brake pads, actuators, and/or any other suitable componentry configured to assist in decelerating the AV **102**. The steering system **134** can include suitable componentry configured to control the direction of movement of the AV **102** during navigation. The safety system **136** can include lights and signal indicators, a parking brake, airbags, and so forth. The cabin system **138** can include cabin temperature control systems, in-cabin entertainment systems, and so forth. In some examples, the AV **102** might not include human driver actuators (e.g., steering wheel, handbrake, foot brake pedal, foot accelerator pedal, turn signal lever, window wipers, etc.) for controlling the AV **102**. Instead, the cabin system **138** can include one or more client interfaces (e.g., Graphical User Interfaces (GUIs), Voice User Interfaces (VUIs), etc.) for controlling certain aspects of the mechanical systems **130-138**.

(17) The AV **102** can include a local computing device **110** that is in communication with the sensor systems **104-108**, the mechanical systems **130-138**, the data center **150**, and the client computing device **170**, among other systems. The local computing device **110** can include one or more processors and memory, including instructions that can be executed by the one or more processors. The instructions can make up one or more software stacks or components responsible for controlling the AV **102**; communicating with the data center **150**, the client computing device **170**, and other systems; receiving inputs from riders, passengers, and other entities within the AV's environment; logging metrics collected by the sensor systems **104-108**; and so forth. In this example, the local computing device **110** includes a perception stack **112**, a localization stack **114**, a prediction stack **116**, a planning stack **118**, a communications stack **120**, a control stack **122**, an AV operational database **124**, and an HD geospatial database **126**, among other stacks and systems.

(18) The perception stack **112** can enable the AV **102** to “see” (e.g., via cameras, LIDAR sensors, infrared sensors, etc.), “hear” (e.g., via microphones, ultrasonic sensors, RADAR, etc.), and “feel” (e.g., pressure sensors, force sensors, impact sensors, etc.) its environment using information from

the sensor systems **104-108**, the localization stack **114**, the HD geospatial database **126**, other components of the AV, and other data sources (e.g., the data center **150**, the client computing device **170**, third party data sources, etc.). The perception stack **112** can detect and classify objects and determine their current locations, speeds, directions, and the like. In addition, the perception stack **112** can determine the free space around the AV **102** (e.g., to maintain a safe distance from other objects, change lanes, park the AV, etc.). The perception stack **112** can identify environmental uncertainties, such as where to look for moving objects, flag areas that may be obscured or blocked from view, and so forth. In some examples, an output of the prediction stack can be a bounding area around a perceived object that can be associated with a semantic label that identifies the type of object that is within the bounding area, the kinematic of the object (information about its movement), a tracked path of the object, and a description of the pose of the object (its orientation or heading, etc.).

(19) The localization stack **114** can determine the AV's position and orientation (pose) using different methods from multiple systems (e.g., GPS, IMUs, cameras, LIDAR, RADAR, ultrasonic sensors, the HD geospatial database **126**, etc.). For example, in some cases, the AV **102** can compare sensor data captured in real-time by the sensor systems **104-108** to data in the HD geospatial database **126** to determine its precise (e.g., accurate to the order of a few centimeters or less) position and orientation. The AV **102** can focus its search based on sensor data from one or more first sensor systems (e.g., GPS) by matching sensor data from one or more second sensor systems (e.g., LIDAR). If the mapping and localization information from one system is unavailable, the AV **102** can use mapping and localization information from a redundant system and/or from remote data sources.

(20) The prediction stack **116** can receive information from the localization stack **114** and objects identified by the perception stack **112** and predict a future path for the objects. In some examples, the prediction stack **116** can output several likely paths that an object is predicted to take along with a probability associated with each path. For each predicted path, the prediction stack **116** can also output a range of points along the path corresponding to a predicted location of the object along the path at future time intervals along with an expected error value for each of the points that indicates a probabilistic deviation from that point.

(21) The planning stack **118** can determine how to maneuver or operate the AV **102** safely and efficiently in its environment. For example, the planning stack **118** can receive the location, speed, and direction of the AV **102**, geospatial data, data regarding objects sharing the road with the AV **102** (e.g., pedestrians, bicycles, vehicles, ambulances, buses, cable cars, trains, traffic lights, lanes, road markings, etc.) or certain events occurring during a trip (e.g., emergency vehicle blaring a siren, intersections, occluded areas, street closures for construction or street repairs, double-parked cars, etc.), traffic rules and other safety standards or practices for the road, user input, and other relevant data for directing the AV **102** from one point to another and outputs from the perception stack **112**, localization stack **114**, and prediction stack **116**. The planning stack **118** can determine multiple sets of one or more mechanical operations that the AV **102** can perform (e.g., go straight at a specified rate of acceleration, including maintaining the same speed or decelerating; turn on the left blinker, decelerate if the AV is above a threshold range for turning, and turn left; turn on the right blinker, accelerate if the AV is stopped or below the threshold range for turning, and turn right; decelerate until completely stopped and reverse; etc.), and select the best one to meet changing road conditions and events. If something unexpected happens, the planning stack **118** can select from multiple backup plans to carry out. For example, while preparing to change lanes to turn right at an intersection, another vehicle may aggressively cut into the destination lane, making the lane change unsafe. The planning stack **118** could have already determined an alternative plan for such an event. Upon its occurrence, it could help direct the AV **102** to go around the block instead of blocking a current lane while waiting for an opening to change lanes.

(22) The control stack **122** can manage the operation of the vehicle propulsion system **130**, the

braking system **132**, the steering system **134**, the safety system **136**, and the cabin system **138**. The control stack **122** can receive sensor signals from the sensor systems **104-108** as well as communicate with other stacks or components of the local computing device **110** or a remote system (e.g., the data center **150**) to effectuate operation of the AV **102**. For example, the control stack **122** can implement the final path or actions from the multiple paths or actions provided by the planning stack **118**. This can involve turning the routes and decisions from the planning stack **118** into commands for the actuators that control the AV's steering, throttle, brake, and drive unit.

(23) The communications stack **120** can transmit and receive signals between the various stacks and other components of the AV **102** and between the AV **102**, the data center **150**, the client computing device **170**, and other remote systems. The communications stack **120** can enable the local computing device **110** to exchange information remotely over a network, such as through an antenna array or interface that can provide a metropolitan WIFI network connection, a mobile or cellular network connection (e.g., Third Generation (3G), Fourth Generation (4G), Long-Term Evolution (LTE), 5th Generation (5G), etc.), and/or other wireless network connection (e.g., License Assisted Access (LAA), Citizens Broadband Radio Service (CBRS), MULTEFIRE, etc.). The communications stack **120** can also facilitate the local exchange of information, such as through a wired connection (e.g., a user's mobile computing device docked in an in-car docking station or connected via Universal Serial Bus (USB), etc.) or a local wireless connection (e.g., Wireless Local Area Network (WLAN), Bluetooth®, infrared, etc.).

(24) The HD geospatial database **126** can store HD maps and related data of the streets upon which the AV **102** travels. In some examples, the HD maps and related data can comprise multiple layers, such as an areas layer, a lanes and boundaries layer, an intersections layer, a traffic controls layer, and so forth. The areas layer can include geospatial information indicating geographic areas that are drivable (e.g., roads, parking areas, shoulders, etc.) or not drivable (e.g., medians, sidewalks, buildings, etc.), drivable areas that constitute links or connections (e.g., drivable areas that form the same road) versus intersections (e.g., drivable areas where two or more roads intersect), and so on. The lanes and boundaries layer can include geospatial information of road lanes (e.g., lane centerline, lane boundaries, type of lane boundaries, etc.) and related attributes (e.g., direction of travel, speed limit, lane type, etc.). The lanes and boundaries layer can also include three-dimensional (3D) attributes related to lanes (e.g., slope, elevation, curvature, etc.). The intersections layer can include geospatial information of intersections (e.g., crosswalks, stop lines, turning lane centerlines and/or boundaries, etc.) and related attributes (e.g., permissive, protected/permissive, or protected only left turn lanes; legal or illegal u-turn lanes; permissive or protected only right turn lanes; etc.). The traffic controls lane can include geospatial information of traffic signal lights, traffic signs, and other road objects and related attributes.

(25) The AV operational database **124** can store raw AV data generated by the sensor systems **104-108**, stacks **112-122**, and other components of the AV **102** and/or data received by the AV **102** from remote systems (e.g., the data center **150**, the client computing device **170**, etc.). In some examples, the raw AV data can include HD LIDAR point cloud data, image data, RADAR data, GPS data, and other sensor data that the data center **150** can use for creating or updating AV geospatial data or for creating simulations of situations encountered by AV **102** for future testing or training of various machine learning algorithms that are incorporated in the local computing device **110**.

(26) The data center **150** can include a private cloud (e.g., an enterprise network, a co-location provider network, etc.), a public cloud (e.g., an Infrastructure as a Service (IaaS) network, a Platform as a Service (PaaS) network, a Software as a Service (SaaS) network, or other Cloud Service Provider (CSP) network), a hybrid cloud, a multi-cloud, and/or any other network. The data center **150** can include one or more computing devices remote to the local computing device **110** for managing a fleet of AVs and AV-related services. For example, in addition to managing the AV **102**, the data center **150** may also support a ridehailing service (e.g., a ridesharing service), a delivery service, a remote/roadside assistance service, street services (e.g., street mapping, street

patrol, street cleaning, street metering, parking reservation, etc.), and the like.

(27) The data center **150** can send and receive various signals to and from the AV **102** and the client computing device **170**. These signals can include sensor data captured by the sensor systems **104-108**, roadside assistance requests, software updates, ridehailing/ridesharing pick-up and drop-off instructions, and so forth. In this example, the data center **150** includes a data management platform **152**, an Artificial Intelligence/Machine Learning (AI/ML) platform **154**, a simulation platform **156**, a remote assistance platform **158**, and a ridehailing platform **160**, and a map management platform **162**, among other systems.

(28) The data management platform **152** can be a “big data” system capable of receiving and transmitting data at high velocities (e.g., near real-time or real-time), processing a large variety of data and storing large volumes of data (e.g., terabytes, petabytes, or more of data). The varieties of data can include data having different structures (e.g., structured, semi-structured, unstructured, etc.), data of different types (e.g., sensor data, mechanical system data, ridehailing/ridesharing service, map data, audio, video, etc.), data associated with different types of data stores (e.g., relational databases, key-value stores, document databases, graph databases, column-family databases, data analytic stores, search engine databases, time series databases, object stores, file systems, etc.), data originating from different sources (e.g., AVs, enterprise systems, social networks, etc.), data having different rates of change (e.g., batch, streaming, etc.), and/or data having other characteristics. The various platforms and systems of the data center **150** can access data stored by the data management platform **152** to provide their respective services.

(29) The AI/ML platform **154** can provide the infrastructure for training and evaluating machine learning algorithms for operating the AV **102**, the simulation platform **156**, the remote assistance platform **158**, the ridehailing platform **160**, the map management platform **162**, and other platforms and systems. Using the AI/ML platform **154**, data scientists can prepare data sets from the data management platform **152**; select, design, and train machine learning models; evaluate, refine, and deploy the models; maintain, monitor, and retrain the models; and so on.

(30) The simulation platform **156** can enable testing and validation of the algorithms, machine learning models, neural networks, and other development efforts for the AV **102**, the remote assistance platform **158**, the ridehailing platform **160**, the map management platform **162**, and other platforms and systems. The simulation platform **156** can replicate a variety of driving environments and/or reproduce real-world scenarios from data captured by the AV **102**, including rendering geospatial information and road infrastructure (e.g., streets, lanes, crosswalks, traffic lights, stop signs, etc.) obtained from a cartography platform (e.g., map management platform **162**); modeling the behavior of other vehicles, bicycles, pedestrians, and other dynamic elements; simulating inclement weather conditions, different traffic scenarios; and so on.

(31) The remote assistance platform **158** can generate and transmit instructions regarding the operation of the AV **102**. For example, in response to an output of the AI/ML platform **154** or other system of the data center **150**, the remote assistance platform **158** can prepare instructions for one or more stacks or other components of the AV **102**.

(32) The ridehailing platform **160** can interact with a customer of a ridehailing service (e.g., a ridesharing service) via a ridehailing application **172** executing on the client computing device **170**. The client computing device **170** can be any type of computing system such as, for example and without limitation, a server, desktop computer, laptop computer, tablet computer, smartphone, smart wearable device (e.g., smartwatch, smart eyeglasses or other Head-Mounted Display (HMD), smart ear pods, or other smart in-ear, on-ear, or over-ear device, etc.), gaming system, or any other computing device for accessing the ridehailing application **172**. The client computing device **170** can be a customer's mobile computing device or a computing device integrated with the AV **102** (e.g., the local computing device **110**). The ridehailing platform **160** can receive requests to pick up or drop off from the ridehailing application **172** and dispatch the AV **1102** for the trip.

(33) Map management platform **162** can provide a set of tools for the manipulation and

management of geographic and spatial (geospatial) and related attribute data. The data management platform **152** can receive LIDAR point cloud data, image data (e.g., still image, video, etc.), RADAR data, GPS data, and other sensor data (e.g., raw data) from one or more AVs **102**, Unmanned Aerial Vehicles (UAVs), satellites, third-party mapping services, and other sources of geospatially referenced data. The raw data can be processed, and map management platform **162** can render base representations (e.g., tiles (2D), bounding volumes (3D), etc.) of the AV geospatial data to enable users to view, query, label, edit, and otherwise interact with the data. Map management platform **162** can manage workflows and tasks for operating on the AV geospatial data. Map management platform **162** can control access to the AV geospatial data, including granting or limiting access to the AV geospatial data based on user-based, role-based, group-based, task-based, and other attribute-based access control mechanisms. Map management platform **162** can provide version control for the AV geospatial data, such as to track specific changes that (human or machine) map editors have made to the data and to revert changes when necessary. Map management platform **162** can administer release management of the AV geospatial data, including distributing suitable iterations of the data to different users, computing devices, AVs, and other consumers of HD maps. Map management platform **162** can provide analytics regarding the AV geospatial data and related data, such as to generate insights relating to the throughput and quality of mapping tasks.

(34) In some embodiments, the map viewing services of map management platform **162** can be modularized and deployed as part of one or more of the platforms and systems of the data center **150**. For example, the AI/ML platform **154** may incorporate the map viewing services for visualizing the effectiveness of various object detection or object classification models, the simulation platform **156** may incorporate the map viewing services for recreating and visualizing certain driving scenarios, the remote assistance platform **158** may incorporate the map viewing services for replaying traffic incidents to facilitate and coordinate aid, the ridehailing platform **160** may incorporate the map viewing services into the ridehailing application **172** (e.g., client application) to enable passengers to view the AV **102** in transit en route to a pick-up or drop-off location, and so on.

(35) While the autonomous vehicle **102**, the local computing device **110**, and the AV environment **100** are shown to include certain systems and components, one of ordinary skill will appreciate that the autonomous vehicle **102**, the local computing device **110**, and/or the AV environment **100** can include more or fewer systems and/or components than those shown in FIG. 1. For example, the autonomous vehicle **102** can include other services than those shown in FIG. 1 and the local computing device **110** can also include, in some instances, one or more memory devices (e.g., RAM, ROM, cache, and/or the like), one or more network interfaces (e.g., wired and/or wireless communications interfaces and the like), and/or other hardware or processing devices that are not shown in FIG. 1. An illustrative example of a computing device and hardware components that can be implemented with the local computing device **110** is described below with respect to FIG. 5.

(36) FIG. 2 illustrates a diagram that includes an AV (e.g., AV **102**) detecting a flood condition. As noted previously, while AV **102** navigates on a road, various sensors (e.g., sensor systems **104-108** as illustrated in FIG. 1) of AV **102** can capture and collect sensor data that can be used to measure, describe, and/or depict one or more aspects of an environment around AV **102**. More specifically, the sensor data can include measurements associated with a level of precipitation (e.g., a water level or an amount of precipitation) on the ground such as on a road, a street, a highway, and/or any other surface on which AV **102** may operate. For example, a RADAR sensor can produce detailed precipitation information by measuring radar waves that bounce off of precipitation. Examples of sensors can further include a camera, a LiDAR, an ambient light sensor, an infrared sensor, a rain sensor, a rain gauge, an ultrasonic sensor, and so forth. Those skilled in the art will recognize that the systems and techniques described herein may be implemented using a different number of sensors that are mounted on AV **102** as well as different types and/or different positions of sensors

that may be configured to detect a precipitation condition and/or measure a level of precipitation on a road in real-time.

(37) In some examples, the systems and techniques described herein can, based on the sensor data, determine a level of precipitation and/or a progression rate of precipitation in one or more locations or spots around AV **102**. In the illustrative example of FIG. 2, as AV **102** navigates on a road, one or more sensors of AV **102** (e.g., sensor systems **104-108**) can detect a precipitation condition (e.g., rain, drizzle, hail, snow, etc.) and measure the level of precipitation on the road (e.g., ground surface **202**). In some examples, AV **102** can identify one or more locations or spots on the road where the level of precipitation is higher than a precipitation threshold. For example, AV **102** can detect excess water on a street gutter **210** below a curb on the left side of the road. In another example, AV **102** can detect the level of precipitation in one or more potholes (e.g., potholes **220**). In some aspects, if a level or amount of precipitation on the road (e.g., ground surface **202**), on street gutter **210**, and/or potholes **220** exceeds a predetermined precipitation threshold, the systems and techniques described herein can identify street gutter **210** and potholes **220** as a location affected by or in a flood condition.

(38) In some examples, based on the detection of one or more spots that have a level of precipitation higher than a precipitation threshold, the systems and techniques described herein can predict and detect a potential, imminent, or occurring flood condition at certain locations and/or spots. Also, rather than estimating or measuring rainfalls at one place and generalizing the flood condition for a nearby area, the systems and techniques of the present technology can identify a location, spot, or portion of a road segment at a small scale with detailed predictions and detection. For example, AV **102** can be routed to a lane that may be drivable and has a low level of precipitation (e.g., level of precipitation is below a threshold value) and avoid a lane that has a spot or spots (e.g., potholes **220**) with higher levels of precipitation rather than making a detour to avoid the entire area.

(39) FIG. 3 illustrates an example system **300** for routing and/or dispatching an AV in a flood condition. In some aspects, system **300** may include AV **102**, which comprises one or more sensors (e.g., sensor systems **104-108**) that generate sensor data. In some examples, system **300** may include data center **310** (similar to data center **150** as illustrated in FIG. 1), which can store various types of data including the sensor data captured by the sensor(s) of AV **102**. In some aspects, system **300** may include fleet management system **320**, which can manage the routing and/or dispatching of one or more AVs in a fleet (e.g., AV **102**). In some examples, system **300** can be part of the internal computing system of AV **102** (similar to local computing device **110** as illustrated in FIG. 1), remote computing system, or a combination thereof.

(40) In some examples, data center **310** is configured to collect and store data from various data sources that can help predict and determine a potential, imminent, or occurring flood event/condition. In some cases, data center **310** can receive and/or store sensor data that is captured by the sensor(s) of AV **102** (e.g., sensor systems **104-108** such as a camera, a LiDAR, a RADAR, a rain gauge, a rain sensor, etc.). As described in connection with FIG. 2, the sensor data can include measurements associated with a level of precipitation (e.g., a water level or an amount of precipitation) on a road.

(41) In some examples, data center **310** can receive fleet data from fleet management system **320**. More specifically, fleet management system **320** can provide sensor data that is captured by one or more AVs in a fleet that are navigating on road. The fleet sensor data can enable real-time data collection associated with a flood condition that covers a geographic area on the road.

(42) In some cases, data center **310** can include map data that is descriptive of road geometries, road elements, and/or environmental elements. In some examples, the map data (e.g., a high-definition map) can include details of the road geometries (e.g., 3-dimensional geometry of the road, road shape, etc.), road elements (e.g., semantic information or labels of road elements, shapes of the road elements in 3-D, etc.), and/or any other map data. In some examples, the map data is

descriptive of an elevation of ground (e.g., z-levels) that reflects where a higher or lower ground is on a map. In some cases, the map data relating to the elevation of ground can indicate absolute elevation. For example, one geographical area or road/road segment can be higher in elevation than an adjacent area, which may indicate how excess precipitation would flow (e.g., the level of predicted precipitation). In some cases, the map data includes geometric information on road elements such as street gutter, curb, drainage, potholes, etc. For instance, the map data can indicate the height of the curb, the depth of potholes, the size of the drainage, and so on. As follows, when coupled with the measurements associated with a level of precipitation based on the sensor data, the map data can provide the probability of flood (or flood risk) or the progression rate of precipitation at certain locations on a map.

(43) In some examples, data center **310** can include absorbability data that is associated with the capability of the surface of the ground to absorb water. For example, if the ground has a high absorbability and can absorb a lot of water (e.g., rainfalls), then the flood probability can be low and/or the progression rate of precipitation on the ground can be slow. In another example, if the ground has a low absorbability (e.g., ground is saturated) and has a low chance of absorbing water, then the flood probability can be high and/or the progression rate of precipitation on the ground can be high/fast. As follows, the absorbability of the ground coupled with the precipitation level can be used to make a prediction of the precipitation level or water depth (also, the flood probability) on the ground.

(44) In some aspects, data center **310** can include historical data that is indicative of one or more locations, spots, or road segments that are subject to frequent flooding. For example, the historical data can provide zones, areas, locations, spots, or road segments that have been flood-prone in the past. In some cases, data center **310** can further include weather data that is received from third-party weather forecasting organizations (e.g., weather forecasting stations, a National Severe Storms Laboratory (NSSL), etc.).

(45) In some cases, system **300** can leverage a combination of different types of data (e.g., sensor data, fleet data, map data, absorbability data, historical data, and/or third-party weather data as described above) stored in data center **310** to predict and detect a flood condition on road. In some examples, based on a combination of data from various sources, system **300** can determine a progression rate of precipitation (e.g., a rate/pace of precipitation or running water on the surface of the ground) so that more accurate predictions can be made in advance or timely for the safe operation of AV **102**.

(46) In some cases, fleet management system **320** can send signals to AV **102** to notify the potential, imminent, or occurring flood condition based on the combination of data available in data center **310** so that AV **102** can be routed/re-routed to avoid certain locations or specific spots that may be affected by a flood condition.

(47) In some aspects, fleet management system **320** and/or AV **102** can generate and/or provide an updated map based on the combination of available data associated with a potential, imminent, or occurring flood condition. A map can be updated continuously as the data in data center **310** can be collected and monitored in real time. In some examples, the updated map can include visualized markings of safe and/or dangerous zones or spots based on the predictions and detection of the flood condition. For example, the updated map can define a zone or a spot that AV **102** should avoid due to the flood condition. In some aspects, the updated map can provide routing information such as indications of impassible (not drivable) roadways or lanes due to the potential, imminent, or occurring flood conditions so that AV **102** can route/re-route its path accordingly. Further, the updated map can show a progression rate of precipitation on road and/or a flood probability or flood risk, which can be calculated based on the combination of data available in data center **310**.

(48) Further, in some examples, AV **102** can transmit signals to fleet management system **320** so that one or more AVs (not illustrated) in the fleet can be routed/re-routed to avoid certain locations or specific spots that may be affected by a flood condition. In some examples, the signals to fleet

management system **320** can include an updated map that reflects visualized marking of safe or dangerous zones or spots, routing information, a progression rate of precipitation, and/or a flood probability or flood risk as illustrated above.

(49) In some examples, fleet management system **320** and/or data center **310** can, based on the predictions and detection of a flood condition as described above, calibrate or re-calibrate AV **102** to adapt to a potential, imminent, or occurring flood condition. For example, fleet management system **320** can send instructions to AV **102** to adjust a tire pressure, a maximum or minimum speed/acceleration/deceleration rate, braking parameters, steering/turning parameters based on the combination of available data (e.g., the level of precipitation on road, the progression rate of the precipitation/rainfalls, and so on).

(50) In some examples, AV **102** can be associated with a user **302** (e.g., a user with client computing device **170** as illustrated in FIG. **1**). For example, system **300** can include a user interface service that can communicate with user **302**, for example, via ridehailing application **172** as described in connection with FIG. **1**. In some aspects, system **300** (e.g., user interface service) can present visualizations of an updated map that shows potential, imminent, or occurring flood conditions along with a probability of flood risk or a level of precipitation at one or more locations on a map. Further, system **300** (e.g., user interface service) can provide, via ridehailing application **172**, in-app interactions such as warnings/alerts, alternative route suggestions, and updated safe spots in real-time (e.g., during a flood condition).

(51) In some aspects, system **300** (e.g., a ridehailing platform **160** as illustrated in FIG. **1**) can be configured to interact with client computing device **170** (e.g., ridehailing application **172** that is operating on a device of user **302**). In some examples, system **300** can receive requests to be picked up or dropped off and can dispatch an AV (e.g., AV **102** or any other AV) for the requested trip. For example, fleet management system **320** can dispatch an AV in the fleet that is closest to a pick-up location. In some cases, in scheduling a pickup or drop-off or planning the requested trip, system **300** can provide instructions to the dispatched AV to go around any zones, locations, areas, spots, or road segments that are identified to have a potential, imminent, or occurring flood condition. In some aspects, system **300** can determine a pickup or drop-off location, based on the combination of data available in data center **310** (e.g., flood data associated with a flood condition, map data including elevation data, absorbability data, historical data etc.), that minimizes a walking distance between user **302** and the pickup location or between a user destination and the drop-off location to ensure the safety of user **302** in a flood event, for example, based on the level of precipitation.

(52) In some examples, system **300** (e.g., fleet management system **320**) can identify a location of user **302** who might be in danger or expected to be affected by a flood event. In some cases, system **300** may transmit an alert signal to a wireless device or user device that is associated with user **302** to notify user **302** of a potential, imminent, or occurring flood condition. In some aspects, system **300** can dispatch an AV (e.g., AV **102** or any AV) to user **302** so that user **302** can evacuate the affected area.

(53) In some aspects, system **300** can allow user **302** to link other users (e.g., family members) on ridehailing application **172** so that location information can be shared during a potential, imminent, or occurring flood event. For example, an alert signal can be sent to user **302** and other users whose accounts are linked together on ridehailing application **172** when one of them is identified to be located in or near an area, zone, or spot that is affected (or likely to be affected) by a flood event. Further, a location of user **302** and the other users can be shared visually on a map so that one's location can be tracked by one another.

(54) FIG. **4** illustrates a flowchart illustrating an example process **400** for routing an AV in a flood condition. Although the example process **400** depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of process **400**. In other examples, different components of an

example device or system that implements process **400** may perform functions at substantially the same time or in a specific sequence.

(55) At block **410**, process **400** can include collecting sensor data from one or more AVs. In some examples, the sensor data can include measurements associated with a level of precipitation in one or more locations. For example, while AV **102** is navigating on a road, various sensors of AV **102** (e.g., sensor systems **104-108**) can capture and collect sensor data that includes measurements associated with a level of precipitation (e.g., a water level or an amount of precipitation on the surface of the ground).

(56) At block **420**, process **400** can include receiving map data descriptive of an elevation of ground at the one or more locations. More specifically, map data (e.g., a high-definition map) can include 3-D representations of road geometries and road elements that reflect the elevation (in z-level) of the ground. For example, the map data can indicate the size of street gutter **210**, the height of the curb above street gutter **210**, or the size of potholes **220** as illustrated in FIG. 2. Such information, when coupled with the measurements associated with a level of precipitation based on the sensor data at block **410**, can be used to determine a probability of flood (or flood risk) and/or a progression rate of precipitation.

(57) At block **430**, process **400** can include determining that the level of precipitation in the one or more locations relative to the elevation of the ground at the one or more locations is higher than a threshold. For example, system **300** can compare the level of precipitation relative to the elevation of ground (e.g., an amount of water or a level of precipitation in potholes **220**) against a predetermined precipitation threshold. Further, system **300** can identify one or more locations in which the level of precipitation relative to the ground elevation is higher than a threshold as an area/location in a flood condition.

(58) In some examples, process **400** can include receiving absorbability data indicative of an absorbability of ground from water in the one or more locations, historical data indicative of locations, spots, or road segments that are subject to frequent flooding, third-party weather data providing a status of flood event, and fleet sensor data collected by one or more sensors of AVs in the same fleet as illustrated in connection with FIG. 3.

(59) In some aspects, process **400** can include determining a progression rate of precipitation based on a combination of the sensor data from the one or more AVs, the elevation of ground at the one or more locations, the absorbability data, and the historical data. For example, system **300** can determine a progression rate of precipitation based on a combination of data available in data center **310** including sensor data from AV **102**, fleet data from fleet management system **320**, map data, historical data, third-party weather data, and any applicable data associated with a flood condition on the road.

(60) At block **440**, process **400** can include transmitting an alert signal to the one or more AVs to notify a flood condition at the one or more locations. For example, fleet management system **320** can transmit an alert signal to AV **102** to notify a flood condition in an environment near AV **102** so AV **102** can route/re-route to avoid the locations or spots that are in a flood condition.

(61) FIG. 5 illustrates an example processor-based system with which some aspects of the subject technology can be implemented. For example, processor-based system **500** can be any computing device making up, or any component thereof in which the components of the system are in communication with each other using connection **505**. Connection **505** can be a physical connection via a bus, or a direct connection into processor **510**, such as in a chipset architecture. Connection **505** can also be a virtual connection, networked connection, or logical connection.

(62) In some embodiments, computing system **500** is a distributed system in which the functions described in this disclosure can be distributed within a datacenter, multiple data centers, a peer network, etc. In some embodiments, one or more of the described system components represents many such components each performing some or all of the function for which the component is described. In some embodiments, the components can be physical or virtual devices.

(63) Example system **500** includes at least one processing unit (Central Processing Unit (CPU) or processor) **510** and connection **505** that couples various system components including system memory **515**, such as Read-Only Memory (ROM) **520** and Random-Access Memory (RAM) **525** to processor **510**. Computing system **500** can include a cache of high-speed memory **512** connected directly with, in close proximity to, or integrated as part of processor **510**.

(64) Processor **510** can include any general-purpose processor and a hardware service or software service, such as services **532**, **534**, and **536** stored in storage device **530**, configured to control processor **510** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **510** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

(65) To enable user interaction, computing system **500** includes an input device **545**, which can represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech, etc. Computing system **500** can also include output device **535**, which can be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems can enable a user to provide multiple types of input/output to communicate with computing system **500**. Computing system **500** can include communication interface **540**, which can generally govern and manage the user input and system output. The communication interface may perform or facilitate receipt and/or transmission wired or wireless communications via wired and/or wireless transceivers, including those making use of an audio jack/plug, a microphone jack/plug, a Universal Serial Bus (USB) port/plug, an Apple® Lightning® port/plug, an Ethernet port/plug, a fiber optic port/plug, a proprietary wired port/plug, a BLUETOOTH® wireless signal transfer, a BLUETOOTH® low energy (BLE) wireless signal transfer, an IBEACON® wireless signal transfer, a Radio-Frequency Identification (RFID) wireless signal transfer, Near-Field Communications (NFC) wireless signal transfer, Dedicated Short Range Communication (DSRC) wireless signal transfer, 802.11 Wi-Fi® wireless signal transfer, Wireless Local Area Network (WLAN) signal transfer, Visible Light Communication (VLC) signal transfer, Worldwide Interoperability for Microwave Access (WiMAX), Infrared (IR) communication wireless signal transfer, Public Switched Telephone Network (PSTN) signal transfer, Integrated Services Digital Network (ISDN) signal transfer, 3G/4G/5G/LTE cellular data network wireless signal transfer, ad-hoc network signal transfer, radio wave signal transfer, microwave signal transfer, infrared signal transfer, visible light signal transfer signal transfer, ultraviolet light signal transfer, wireless signal transfer along the electromagnetic spectrum, or some combination thereof.

(66) Communication interface **540** may also include one or more Global Navigation Satellite System (GNSS) receivers or transceivers that are used to determine a location of the computing system **500** based on receipt of one or more signals from one or more satellites associated with one or more GNSS systems. GNSS systems include, but are not limited to, the US-based Global Positioning System (GPS), the Russia-based Global Navigation Satellite System (GLONASS), the China-based BeiDou Navigation Satellite System (BDS), and the Europe-based Galileo GNSS. There is no restriction on operating on any particular hardware arrangement, and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

(67) Storage device **530** can be a non-volatile and/or non-transitory and/or computer-readable memory device and can be a hard disk or other types of computer readable media which can store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, a floppy disk, a flexible disk, a hard disk, magnetic tape, a magnetic strip/stripe, any other magnetic storage medium, flash memory, memristor memory, any other solid-state memory, a Compact Disc (CD) Read Only Memory (CD-ROM) optical disc, a rewritable CD optical disc, a Digital Video Disk (DVD) optical disc, a Blu-

ray Disc (BD) optical disc, a holographic optical disc, another optical medium, a Secure Digital (SD) card, a micro SD (microSD) card, a Memory Stick® card, a smartcard chip, a EMV chip, a Subscriber Identity Module (SIM) card, a mini/micro/nano/pico SIM card, another Integrated Circuit (IC) chip/card, Random-Access Memory (RAM), Atatic RAM (SRAM), Dynamic RAM (DRAM), Read-Only Memory (ROM), Programmable ROM (PROM), Erasable PROM (EPROM), Electrically Erasable PROM (EEPROM), flash EPROM (FLASH EPROM), cache memory (L1/L2/L3/L4/L5/L#), Resistive RAM (RRAM/ReRAM), Phase Change Memory (PCM), Spin Transfer Torque RAM (STT-RAM), another memory chip or cartridge, and/or a combination thereof.

(68) Storage device **530** can include software services, servers, services, etc., that when the code that defines such software is executed by the processor **510**, it causes the system **500** to perform a function. In some embodiments, a hardware service that performs a particular function can include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **510**, connection **505**, output device **535**, etc., to carry out the function.

(69) Embodiments within the scope of the present disclosure may also include tangible and/or non-transitory computer-readable storage media or devices for carrying or having computer-executable instructions or data structures stored thereon. Such tangible computer-readable storage devices can be any available device that can be accessed by a general purpose or special purpose computer, including the functional design of any special purpose processor as described above. By way of example, and not limitation, such tangible computer-readable devices can include RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other device which can be used to carry or store desired program code in the form of computer-executable instructions, data structures, or processor chip design. When information or instructions are provided via a network or another communications connection (either hardwired, wireless, or combination thereof) to a computer, the computer properly views the connection as a computer-readable medium. Thus, any such connection is properly termed a computer-readable medium. Combinations of the above should also be included within the scope of the computer-readable storage devices.

(70) Computer-executable instructions include, for example, instructions and data which cause a general-purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. Computer-executable instructions also include program modules that are executed by computers in stand-alone or network environments. Generally, program modules include routines, programs, components, data structures, objects, and the functions inherent in the design of special-purpose processors, etc. that perform tasks or implement abstract data types. Computer-executable instructions, associated data structures, and program modules represent examples of the program code means for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described in such steps.

(71) Other embodiments of the disclosure may be practiced in network computing environments with many types of computer system configurations, including personal computers, hand-held devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network Personal Computers (PCs), minicomputers, mainframe computers, and the like.

Embodiments may also be practiced in distributed computing environments where tasks are performed by local and remote processing devices that are linked (either by hardwired links, wireless links, or by a combination thereof) through a communications network. In a distributed computing environment, program modules may be located in both local and remote memory storage devices.

(72) The various embodiments described above are provided by way of illustration only and should

not be construed to limit the scope of the disclosure. For example, the principles herein apply equally to optimization as well as general improvements. Various modifications and changes may be made to the principles described herein without following the example embodiments and applications illustrated and described herein, and without departing from the spirit and scope of the disclosure.

(73) Claim language or other language in the disclosure reciting “at least one of” a set and/or “one or more” of a set indicates that one member of the set or multiple members of the set (in any combination) satisfy the claim. For example, claim language reciting “at least one of A and B” or “at least one of A or B” means A, B, or A and B. In another example, claim language reciting “at least one of A, B, and C” or “at least one of A, B, or C” means A, B, C, or A and B, or A and C, or B and C, or A and B and C. The language “at least one of” a set and/or “one or more” of a set does not limit the set to the items listed in the set. For example, claim language reciting “at least one of A and B” or “at least one of A or B” can mean A, B, or A and B, and can additionally include items not listed in the set of A and B.

(74) Illustrative examples of the disclosure include: Aspect 1. A system comprising: a memory; and one or more processors coupled to the memory, wherein the one or more processors are configured to: collect sensor data from one or more autonomous vehicles (AVs), wherein the sensor data includes measurements associated with a level of precipitation in one or more locations; receive map data descriptive of an elevation of ground at the one or more locations; determine that the level of precipitation in the one or more locations relative to the elevation of ground at the one or more locations is higher than a threshold; and transmit an alert signal to the one or more AVs, the alert signal notifying of a flood condition at the one or more locations. Aspect 2. The system of Aspect 1, wherein the alert signal includes a visualization of the flood condition at the one or more locations on a map. Aspect 3. The system of Aspect 1 or 2, wherein the one or more processors are further configured to: receive absorbability data indicative of an absorbability of ground from water in the one or more locations. Aspect 4. The system of Aspect 3, wherein the one or more processors are further configured to: determine a progression rate of precipitation based on the sensor data from the one or more AVs, the elevation of ground at the one or more locations, and the absorbability data. Aspect 5. The system of any of Aspects 1 to 4, wherein the one or more processors are further configured to: identify a user associated with at least one AV of the one or more AVs, wherein the user is located within a threshold distance of at least one location of the one or more locations; and dispatch the at least one AV to pick up the user. Aspect 6. The system of any of Aspects 1 to 5, wherein at least one of the one or more AVs is associated with a ridehail service, and the one or more processors are further configured to: determine a pickup location or a drop-off location that minimizes a walking distance between a user and the pickup location or between a user destination and the drop-off location based on the level of precipitation. Aspect 7. The system of any of Aspects 1 to 6, wherein the sensor data is captured by a Radio Detection and Ranging (RADAR) sensor of the one or more AVs. Aspect 8. A method comprising: collecting sensor data from one or more autonomous vehicles (AVs), wherein the sensor data includes measurements associated with a level of precipitation in one or more locations; receiving map data descriptive of an elevation of ground at the one or more locations; determining that the level of precipitation in the one or more locations relative to the elevation of ground at the one or more locations is higher than a threshold; and transmitting an alert signal to the one or more AVs, the alert signal notifying of a flood condition at the one or more locations. Aspect 9. The method of Aspect 8, wherein the alert signal includes a visualization of the flood condition at the one or more locations on a map. Aspect 10. The method of Aspect 8 or 9, further comprising: receiving absorbability data indicative of an absorbability of ground from water in the one or more locations. Aspect 11. The method of Aspect 10, further comprising: determining a progression rate of precipitation based on the sensor data from the one or more AVs, the elevation of ground at the one or more locations, and the absorbability data. Aspect 12. The method of any of Aspects 8-11, further comprising: identifying a

user associated with at least one AV of the one or more AVs, wherein the user is located within a threshold distance of at least one location of the one or more locations; and dispatching the at least one AV to pick up the user. Aspect 13. The method of any of Aspects 8-12, wherein at least one of the one or more AVs is associated with a ridehail service, and further comprising: determining a pickup location or a drop-off location that minimizes a walking distance between a user and the pickup location or between a user destination and the drop-off location based on the level of precipitation. Aspect 14. The method of any of Aspects 8-13, wherein the sensor data is captured by a Radio Detection and Ranging (RADAR) sensor of the one or more AVs. Aspect 15. A non-transitory computer-readable medium having stored thereon instructions which, when executed by one or more processors, cause the one or more processors to: collect sensor data from one or more autonomous vehicles (AVs), wherein the sensor data includes measurements associated with a level of precipitation in one or more locations; receive map data descriptive of an elevation of ground at the one or more locations; determine that the level of precipitation in the one or more locations relative to the elevation of ground at the one or more locations is higher than a threshold; and transmit an alert signal to the one or more AVs, the alert signal notifying of a flood condition at the one or more locations. Aspect 16. The non-transitory computer-readable medium of Aspect 15, wherein the alert signal includes a visualization of the flood condition at the one or more locations on a map. Aspect 17. The non-transitory computer-readable medium of Aspects 15 or 16, comprising further instructions configured to cause the computer or the processor to: receive absorbability data indicative of an absorbability of ground from water in the one or more locations. Aspect 18. The non-transitory computer-readable medium of Aspect 17, comprising further instructions configured to cause the computer or the processor to: determine a progression rate of precipitation based on the sensor data from the one or more AVs, the elevation of ground at the one or more locations, and the absorbability data. Aspect 19. The non-transitory computer-readable medium of any of Aspects 15-18, comprising further instructions configured to cause the computer or the processor to: identify a user associated with at least one AV of the one or more AVs, wherein the user is located within a threshold distance of at least one location of the one or more locations; and dispatch the at least one AV to pick up the user. Aspect 20. The non-transitory computer-readable medium of any of Aspects 15-19, wherein at least one of the one or more AVs is associated with a ridehail service, and comprising further instructions configured to cause the computer or the processor to: determine a pickup location or a drop-off location that minimizes a walking distance between a user and the pickup location or between a user destination and the drop-off location based on the level of precipitation. Aspect 21. An autonomous vehicle comprising: a memory and one or more processors coupled to the memory, the one or more processors being configured to perform a method according to any of Aspects 8 to 14. Aspect 22. A system comprising means for performing a method according to any of Aspects 8 to 14.

Claims

1. A system comprising: a memory; and one or more processors coupled to the memory, wherein the one or more processors are configured to: collect sensor data from a Radio Detection and Ranging (RADAR) sensor of one or more autonomous vehicles (AVs), wherein the RADAR sensor measures radar waves that reflect off of precipitation, and wherein the sensor data includes measurements associated with a level of precipitation representing a water level on a ground surface at one or more locations; receive map data descriptive of an elevation of ground at the one or more locations; determine that the level of precipitation in the one or more locations relative to the elevation of ground at the one or more locations is higher than a threshold; and transmit an alert signal to the one or more AVs, the alert signal notifying of a flood condition at the one or more locations.
2. The system of claim 1, wherein the alert signal includes a visualization of the flood condition at

the one or more locations on a map.

3. The system of claim 1, wherein the one or more processors are further configured to receive absorbability data indicative of an absorbability of ground from water in the one or more locations.

4. The system of claim 3, wherein the one or more processors are further configured to determine a progression rate of precipitation based on the sensor data from the one or more AVs, the elevation of ground at the one or more locations, and the absorbability data.

5. The system of claim 1, wherein the one or more processors are further configured to: identify a user associated with at least one AV of the one or more AVs, wherein the user is located within a threshold distance of at least one location of the one or more locations; and dispatch the at least one AV to pick up the user.

6. The system of claim 1, wherein at least one of the one or more AVs is associated with a ridehail service, and the one or more processors are further configured to determine a pickup location or a drop-off location that minimizes a walking distance between a user and the pickup location or between a user destination and the drop-off location based on the level of precipitation.

7. A method comprising: collecting sensor data from a Radio Detection and Ranging (RADAR) sensor of one or more autonomous vehicles (AVs), wherein the RADAR sensor measures radar waves that reflect off of precipitation, and wherein the sensor data includes measurements associated with a level of precipitation representing a water level on a ground surface at one or more locations; receiving map data descriptive of an elevation of ground at the one or more locations; determining that the level of precipitation in the one or more locations relative to the elevation of ground at the one or more locations is higher than a threshold; and transmitting an alert signal to the one or more AVs, the alert signal notifying of a flood condition at the one or more locations.

8. The method of claim 7, wherein the alert signal includes a visualization of the flood condition at the one or more locations on a map.

9. The method of claim 7, further comprising receiving absorbability data indicative of an absorbability of ground from water in the one or more locations.

10. The method of claim 9, further comprising determining a progression rate of precipitation based on the sensor data from the one or more AVs, the elevation of ground at the one or more locations, and the absorbability data.

11. The method of claim 7, further comprising: identifying a user associated with at least one AV of the one or more AVs, wherein the user is located within a threshold distance of at least one location of the one or more locations; and dispatching the at least one AV to pick up the user.

12. The method of claim 7, wherein at least one of the one or more AVs is associated with a ridehail service, and further comprising determining a pickup location or a drop-off location that minimizes a walking distance between a user and the pickup location or between a user destination and the drop-off location based on the level of precipitation.

13. A non-transitory computer-readable medium having stored thereon instructions which, when executed by one or more processors, cause the one or more processors to: collect sensor data from a Radio Detection and Ranging (RADAR) sensor of one or more autonomous vehicles (AVs), wherein the RADAR sensor measures radar waves that reflect off of precipitation, and wherein the sensor data includes measurements associated with a level of precipitation representing a water level on a ground surface at one or more locations; receive map data descriptive of an elevation of ground at the one or more locations; determine that the level of precipitation in the one or more locations relative to the elevation of ground at the one or more locations is higher than a threshold; and transmit an alert signal to the one or more AVs, the alert signal notifying of a flood condition at the one or more locations.

14. The non-transitory computer-readable medium of claim 13, wherein the alert signal includes a visualization of the flood condition at the one or more locations on a map.

15. The non-transitory computer-readable medium of claim 13, comprising further instructions

configured to cause the computer or the processor to receive absorbability data indicative of an absorbability of ground from water in the one or more locations.

16. The non-transitory computer-readable medium of claim 15, comprising further instructions configured to cause the computer or the processor to determine a progression rate of precipitation based on the sensor data from the one or more AVs, the elevation of ground at the one or more locations, and the absorbability data.

17. The non-transitory computer-readable medium of claim 13, comprising further instructions configured to cause the computer or the processor to: identify a user associated with at least one AV of the one or more AVs, wherein the user is located within a threshold distance of at least one location of the one or more locations; and dispatch the at least one AV to pick up the user.

18. The non-transitory computer-readable medium of claim 13, wherein at least one of the one or more AVs is associated with a ridehail service, and comprising further instructions configured to cause the computer or the processor to determine a pickup location or a drop-off location that minimizes a walking distance between a user and the pickup location or between a user destination and the drop-off location based on the level of precipitation.
